

Task 1: DNA/Protein Sequence Analysis Report

Protein Analyzed: Interleukin-6 (IL-6) | Tool Used: NCBI BLASTp

Submitted By: Damrudhar Kumar

CodeAlpha Internship Program | June 2026

1. Introduction

1.1 What is BLAST?

BLAST (Basic Local Alignment Search Tool) is one of the most widely used bioinformatics tools for comparing biological sequences. It enables researchers to search nucleotide or protein sequence databases to find regions of similarity between sequences. BLAST calculates the statistical significance of matches, allowing researchers to identify homologous sequences, infer functional relationships, and study evolutionary history.

1.2 Importance of Sequence Analysis

Protein sequence analysis is fundamental to modern biology and medicine. By comparing an unknown protein sequence to known sequences in databases, scientists can:

- Identify the function of newly discovered proteins
- Understand evolutionary relationships between species
- Design targeted drugs based on conserved protein regions
- Diagnose genetic diseases by comparing mutant sequences to normal ones

1.3 Protein of Interest: Interleukin-6 (IL-6)

Interleukin-6 (IL-6) is a multifunctional cytokine — a signaling protein secreted by immune cells. It plays a central role in regulating immune responses, inflammation, and hematopoiesis. IL-6 was selected for this analysis due to its critical biological importance and its well-documented role in diseases such as rheumatoid arthritis, COVID-19, and various cancers. The human IL-6 protein sequence was retrieved from the NCBI Protein Database (Accession: P05231).

2. Methodology

2.1 Sequence Retrieval

The human Interleukin-6 protein sequence was downloaded in FASTA format from the NCBI Protein database (<https://www.ncbi.nlm.nih.gov/protein/>) using the search query 'Interleukin-6 Homo sapiens'. The reviewed entry with accession number P05231 was selected, consisting of 212 amino acids.



Figure 1: NCBI Protein Database homepage showing search query "Interleukin-6 Homo sapiens"

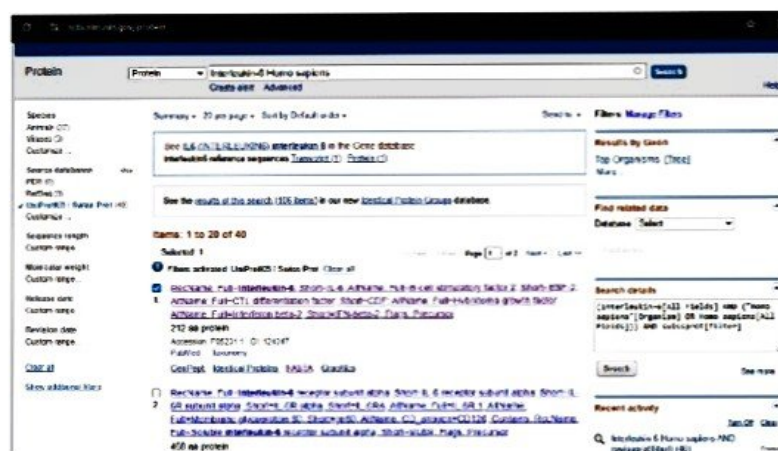


Figure 2: NCBI search results showing Human IL-6 protein entry (UniProtKB/Swiss-Prot: P05231, 212 aa)



Figure 3: FASTA sequence of Human Interleukin-6 retrieved from NCBI (Accession: P05231)

2.2 BLAST Search Parameters

The BLASTp (protein-protein BLAST) tool was used at <https://blast.ncbi.nlm.nih.gov/blastp> with the following parameters:

- Database: Non-redundant protein sequences (nr)

- Algorithm: BLASTp (protein query vs protein database)
- Matrix: BLOSUM62 (default for protein alignments)
- E-value threshold: 0.05
- Maximum target sequences: 100

The screenshot shows the BLASTp input interface. The 'Enter Query Sequence' section has a text area containing a FASTA sequence for IL-6_HUMAN. Below this, there are fields for 'Job Title' and 'Job ID'. The 'Choose Search Set' section shows a dropdown menu for 'Database' set to 'Non-redundant protein sequences (nr)'. There are also fields for 'Organism' and 'Exclude'.

Figure 4: BLASTp input interface with IL-6 FASTA sequence pasted and database set to Non-redundant protein sequences (nr)

2.3 Analysis

Results were analyzed based on three key parameters: percentage identity (how similar the sequences are), E-value (statistical significance of the match), and alignment score (quality of the alignment). Lower E-values and higher percentage identities indicate more biologically meaningful matches.

3.Results

3.1 BLAST Results Summary

The BLASTp search returned a large number of significant hits, indicating that IL-6 is a highly conserved protein across multiple species.

The screenshot shows the BLAST search job summary. The 'Job Title' is 'BLASTp'. The 'Job ID' is 'BLASTP_1860730'. The 'Database' is 'nr'. The 'Query ID' is 'IL6_HUMAN'. The 'Query Length' is 212. The 'Filter Results' section shows the search results with columns for 'Organism', 'Percent Identity', 'E value', and 'Query Coverage'. There are buttons for 'Filter' and 'Reset'.

Figure 5: BLAST search job summary showing query details — IL6_HUMAN, Query Length: 212 amino acids

Description	Accession	Score	E-value
Interleukin-6 isoform 1 precursor	U01428.1	437	3e-154
Interleukin-6 (interferon beta-2), partial	U01428.1	437	3e-154
Interleukin-6 isoform X2	U01428.1	434	2e-153
Interleukin-6 isoform X2	U01428.1	432	1e-152
Interleukin-6 isoform X2	U01428.1	432	1e-152
Interleukin-6	U01428.1	429	2e-151
Interleukin-6	U01428.1	427	1e-150
Interleukin-6 isoform X2	U01428.1	425	1e-149
Interleukin-6	U01428.1	425	1e-149
Interleukin-6 isoform X2	U01428.1	424	3e-149

Figure 6: BLAST results — Descriptions tab showing top hits with E-values and percentage identity scores

❖ The top 10 results are presented in the table below:

S.No.	Description	Organism	Score	E-value	Query Cover	% Identity
1	Interleukin-6 isoform 1 precursor	Homo sapiens (Human)	437	3e-154	100%	100%
2	Interleukin-6 (interferon beta-2), partial	Synthetic construct	437	3e-154	100%	100%
3	Interleukin-6 isoform X2	Pan paniscus (Pygmy chimpanzee)	434	2e-153	100%	99.53%
4	Interleukin-6 isoform X2	Gorilla gorilla gorilla	432	1e-152	100%	99.06%
5	Interleukin-6 isoform X2	Pongo abelii (Sumatran orangutan)	432	1e-152	100%	99.06%
6	Interleukin-6	Homo sapiens (Human isoform)	429	2e-151	100%	99.53%
7	Interleukin-6	Chlorocebus sabaeus (Green monkey)	427	1e-150	100%	98.11%
8	Interleukin-6 isoform X2	Nomascus leucogenys (N. gibbon)	425	1e-149	100%	97.17%
9	Interleukin-6	Papio anubis (Olive baboon)	425	1e-149	100%	97.17%
10	Interleukin-6 isoform X2	Theropithecus gelada (Gelada)	424	3e-149	100%	97.17%

Table 1: Top 10 BLAST hits for Human Interleukin-6 (IL-6)

3.2 Key Observations

Several important patterns emerged from the BLAST results:

- The top two hits showed 100% identity to the query sequence, confirming the accuracy of the search and the presence of redundant database entries for human IL-6.
- Pygmy chimpanzee (*Pan paniscus*) IL-6 showed 99.53% identity, reflecting the very close evolutionary relationship between humans and chimpanzees (~6 million years of divergence).
- Gorilla and Sumatran orangutan IL-6 showed 99.06% identity, consistent with their slightly more distant evolutionary relationship to humans.
- Green monkey (*Chlorocebus sabaeus*) IL-6 showed 98.11% identity, and various Old World monkeys showed ~97% identity, all with extremely low E-values.
- All top 10 hits showed 100% query coverage, meaning the entire IL-6 protein sequence was matched — indicating strong conservation of the full protein across primates.

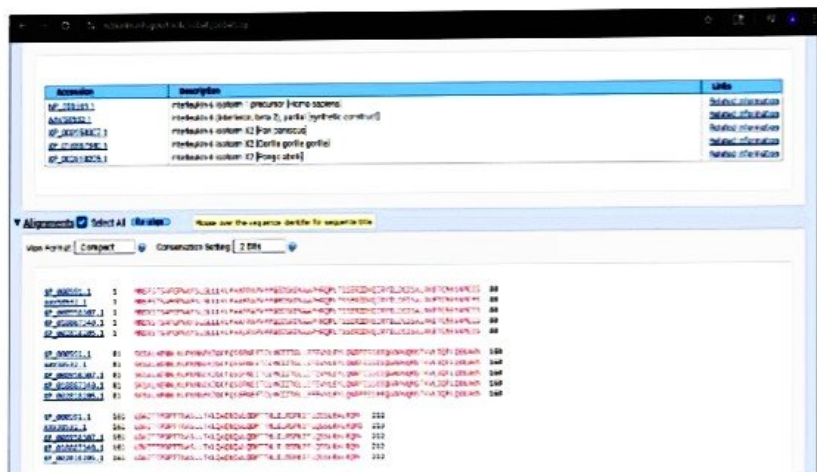


Figure 7: COBALT Multiple Alignment results showing conserved regions of IL-6 across 5 species

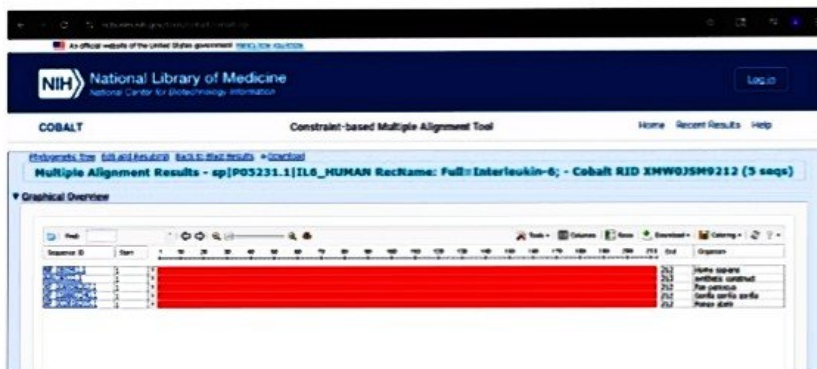


Figure 8: Graphical overview of IL-6 sequence alignment across *Homo sapiens*, *Pan paniscus*, *Gorilla gorilla*, and *Pongo abelii*

3.3 Comparative Analysis

The following table summarizes a comparison between human IL-6 and its orthologues in two other mammalian species:

Parameter	Human vs Chimp	Human vs Mouse
% Identity	99.53%	~65%
E-value	2e-153	~1e-90
Evolutionary distance	Very close (~6 MYA)	Distant (~87 MYA)
Functional similarity	Very high	Moderate

Table 2: Comparative analysis of IL-6 across species

4. Discussion & Interpretation

4.1 Evolutionary Conservation

The extremely high sequence identity of IL-6 across primates (97–100%) strongly suggests that this protein has been under strong purifying (negative) selection throughout evolution. This means that mutations in the

IL-6 gene have been selected against because the protein's function is essential for survival. The high conservation across all primate species in the top BLAST results confirms that IL-6 plays a critical and irreplaceable role in immune regulation.

4.2 Significance of E-values

All top BLAST hits returned E-values in the range of $1e-149$ to $3e-154$. An E-value this close to zero means the probability of finding such a similar sequence by random chance is essentially zero. This confirms that all identified sequences are true biological homologs of human IL-6, not random matches.

4.3 Clinical Relevance

The conservation of IL-6 across species has important clinical implications. IL-6 was found to be massively overproduced in severe COVID-19 patients, contributing to the dangerous 'cytokine storm' that led to organ failure and death. Because of the high conservation between human and primate IL-6, drugs like Tocilizumab (an IL-6 receptor blocker) were developed and successfully used as COVID-19 treatments. Understanding which regions of IL-6 are conserved (and therefore functionally critical) helps researchers design more targeted and effective therapies.

5. Conclusion

This BLAST analysis of human Interleukin-6 (IL-6) successfully demonstrated the power of sequence homology searching in bioinformatics. Key conclusions from this analysis include:

- IL-6 is highly conserved across primate species (97–100% identity), indicating strong evolutionary pressure to maintain its sequence and function.
- All BLAST hits showed extremely low E-values ($3e-149$ to $3e-154$), confirming statistically significant biological homology.
- The 100% query coverage across all top hits demonstrates full-length conservation of the protein.
- The evolutionary pattern of decreasing identity (human > chimp > gorilla > Old World monkeys) is consistent with known primate phylogenetics.
- These findings highlight the importance of IL-6 as a target for immune-related drug development.

This task provided valuable hands-on experience with NCBI protein databases and BLAST analysis tools — fundamental skills for any career in bioinformatics, computational biology, or genomic research.